

# Project Design Phase-II

## Solution Requirements (Functional & Non-functional)

Automated Network Request Management in ServiceNow

| Field         | Details                                            |
|---------------|----------------------------------------------------|
| Date          | 31 January 2025                                    |
| Team ID       | _____                                              |
| Project Name  | Automated Network Request Management in ServiceNow |
| Maximum Marks | 4 Marks                                            |

### Functional Requirements

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                                                                                                                                                                                                                       |
|--------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | Network Request Submission    | <ul style="list-style-type: none"><li>• Access Network Request Service Catalog</li><li>• Select required network request type</li><li>• Enter required request information</li><li>• Submit network service request</li></ul>                                                            |
| FR-2   | Request Form & Validation     | <ul style="list-style-type: none"><li>• Display request-specific fields</li><li>• Mark mandatory fields</li><li>• Validate entered information</li><li>• Prevent submission when required information is missing</li><li>• Support conditional/dynamic fields where applicable</li></ul> |
| FR-3   | Approval Management           | <ul style="list-style-type: none"><li>• Identify required approver</li><li>• Route request for approval</li><li>• Approve or reject request</li><li>• Record approval decision</li><li>• Send approval notifications</li></ul>                                                           |

| FR No. | Functional Requirement (Epic)       | Sub Requirement (Story / Sub-Task)                                                                                                                                                                                                                                                                                |
|--------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-4   | Automated Assignment & Fulfillment  | <ul style="list-style-type: none"> <li>• Identify appropriate network fulfillment group</li> <li>• Automatically assign request/task</li> <li>• Create fulfillment task</li> <li>• Update task status</li> <li>• Allow fulfillment team to record completion</li> </ul>                                           |
| FR-5   | Request Tracking & Notifications    | <ul style="list-style-type: none"> <li>• Display request status</li> <li>• Notify requester about important status changes</li> <li>• Notify approvers about pending approvals</li> <li>• Notify fulfillment team about assigned tasks</li> <li>• Notify requester when request is completed or closed</li> </ul> |
| FR-6   | Request Management & Administration | <ul style="list-style-type: none"> <li>• Allow authorized administrators to manage request categories</li> <li>• Manage assignment groups</li> <li>• Configure approval rules</li> <li>• Maintain request and task information</li> <li>• Monitor pending and completed requests</li> </ul>                       |
| FR-7   | Audit & Reporting                   | <ul style="list-style-type: none"> <li>• Maintain request history</li> <li>• Record approval and fulfillment activities</li> <li>• Maintain audit information</li> <li>• Generate request reports</li> <li>• Provide dashboard-based visibility</li> </ul>                                                        |

## Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement | Description                                                                                                                                                                                                                             |
|--------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1  | Usability                  | The system should provide a simple and user-friendly interface for submitting and tracking network requests. Forms should clearly identify required information and guide users through the request process.                            |
| NFR-2  | Security                   | The system should provide role-based access control and restrict request, approval and administrative functions to authorized users. Security controls such as authentication, authorization, ACLs and audit logging should be applied. |

| FR No. | Non-Functional Requirement | Description                                                                                                                                                                                                                     |
|--------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-3  | Reliability                | The system should reliably process submitted requests, approvals, assignments and notifications without unnecessary data loss or inconsistent status updates. Request history should be maintained throughout the lifecycle.    |
| NFR-4  | Performance                | The system should provide responsive request forms and workflow operations. Forms, validations, queries and automated flows should be designed to avoid unnecessary processing and delays.                                      |
| NFR-5  | Availability               | The system should be accessible to authorized users through the ServiceNow platform during the organization's required service period. Availability depends on the ServiceNow instance and its configured service availability. |
| NFR-6  | Scalability                | The solution should support the addition of new network request types, users, approval rules, fulfillment teams and workflows without requiring a complete redesign of the system.                                              |

## Requirement Traceability Summary

| Requirement Area         | Covered By     |
|--------------------------|----------------|
| Request Submission       | FR-1, FR-2     |
| Approval & Fulfillment   | FR-3, FR-4     |
| Tracking & Communication | FR-5           |
| Administration           | FR-6           |
| Audit & Reporting        | FR-7           |
| Quality Attributes       | NFR-1 to NFR-6 |

The above functional and non-functional requirements define the expected capabilities and quality characteristics of the proposed Automated Network Request Management solution.

These requirements will be used as the basis for subsequent design, development, testing and validation activities.